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1 NOCCARC'S MEDELE OF INDIAN SMART VENTILATORS

With its innovative smart ventilators, Pune-based startup Noccarc is spearheading the 'Make Medical Electronics in India' movement



V310+



V730i

Man proposes, God disposes! Techpreneur and IIT Kanpur alumnus Nikhil Kurele is well-versed with this statement. From securing funding for a startup that built robots for cleaning solar panels to pivoting to smart ventilators during Covid-19, Kurele was driven by his zeal to manufacture a hardware solution in India that addressed a unique yet com-

mon problem for his country.

Eight months after shifting from a 18.6 square metre (200 square feet) room at IIT Kanpur to scale operations to build robots that cleaned solar panels, in June 2019, Kurele, along with his co-founders Harshit Rathore and Tushar Agarwal, watched the pandemic dent their plans. However, the roadblock did not last long, as in May 2020, the

co-founders gathered their knowledge, resources, and funds to address local challenges and introduced ventilators, focusing on comprehensive care.

Identifying the core issues in existing products: multiple patient devices, high running costs, and limited information accessibility to decision-makers, Noccarc developed the V310+ ventilator, which sold more than 2200 units during the second wave of Covid-19 and, by the end of the pandemic, the figure went up to 3600! The company launched a second category of smart ventilators featuring a digital platform that allows remote access to patient information, V730i, in June 2023, and has sold around 55-60 units.

Focused on solving the challenge of data availability and correlating information from various devices, the ventilator comes with a high-end 1.4GHz quad-core ARM processor, with around 21-22 PCB boards employed in different sections. The device has expansion ports, allowing for potential upgrades or additional modules.

Kurele explains that the company wrote and developed their operating system on top of a Linux system while incorporating machine learning to monitor the product's health, allowing automatic alerts if any issues arise. "The accumulated data is used to